

Study Guide

this is a place to start, it is not ALL inclusive

learn the big picture first; then dive into the details; seek understanding not memorization

L15: Digestion

1. Describe anatomy of digestive system: note organs, key cell types, microvilli, glands
2. Describe the processes that accomplish the functions of GI system including motility, secretion, digestion, and absorption
3. Describe how is food broken into small pieces, mixed, and moved in the GI tract?
4. Describe the transport mechanisms for absorption of monomers/subunits of digestion.
5. How is surface area increased in the small intestine? Why is this important?
6. What secretions and enzymes are released by each GI organ? What are their functions? Include all digestive enzymes & their substrates and acid or alkaline fluids secreted and why.
7. Identify the action of the three major GI hormones (gastrin, secretin, CCK)
8. How is the stomach protected from acid? from pepsin?
9. Describe liver functions (*has many*) & exocrine pancreatic functions.
10. Describe the secretion and function of bile.
11. Differentiate pancreatic enzymes from small intestine enzymes.
12. **Potential short answer/essay:**
Trace the complete enzymatic digestion of each macromolecule that we ingest in a meal. Follow its enzymatic digestion and absorption until it lands in the blood/lymph. This should include:
 - a. All enzymatic digestive processes
 - b. The role of each major digestive organ (stomach, small intestine, liver, etc.)
 - c. In what state and how each macromolecule ingested ends up in the blood stream/lymphatics.
 - d. Mechanism by which each monomer/subunit is absorbed across the lumen of the intestines and into the blood or lymphatic capillaries.